Smart Streetlight Controller

GitHub Repo Link:

https://github.com/francisco9192/StreetLightController-FinalProject.git

1. Define Your Objectives

Francisco - My individual objectives are to become very familiar with collaborating on projects through GitHub and VS Code. Another objective of mine is to learn and understand how current modern streetlight controllers work so that I can apply this knowledge to our project and create a functional design.

Kenji - My individual objectives are to improve my C++ programming skills and learn how to create a more complex program using what we learned in class. Additionally, I would like to understand how smart streetlights work and how we can use different inputs like motion detection to automatically control the brightness. I will contribute by helping develop and test the program.

Damilare - My individual objectives will be to apply this code to a pico board or other motherboard through C++ to see how possible it would be to create. A secondary goal is to create a button that can force the light on/off

Team Objective - Our goal is to create a working smart streetlight controller using C++ while improving our programming skills. We also want to make sure everyone contributes equally to finishing the final product.

2. Select Technologies and Tools

Programming language: C++ - We chose C++ because it is what we are learning in class and we want to build our knowledge to create a working program.

Development Tools: Visual Studio Code - We will use VS Code to write and test our program.

Code Management: GitHub - To store and share our code.

Other selected technologies: We plan on using a Pico board because they allow us to easily create a prototype and improve where needed.

3. Create a Project Timeline

Tasks: Automatically On/Off based on surrounding light level. Ability to activate when it detects motion. Possible stretch goal of reading heat.

Milestones: Code able to turn a light on and off. Code able to detect motion. Code able to detect light. Code sensor able to turn light on/off to a specific brightness.

Deadlines: Presentation day is 12/11/2026 from 11:00 AM - 11:14 AM

Task	Deadline	Deliverables
Finish Project Design	End of Oct	Project design
Complete programming tasks	Mid Nov	Working C++ code for each feature
Testing Phase	End of Nov	Tested program with bugs fixed
Finish final report	Dec 8th	Completed report and slideshow
Final Presentation	December 11th	In-class presentation

Team member	Francisco Tovar	Kenji Matsumoto	Damilare Mustapha
Responsibilities	Develop the motion control system	Develop the brightness control system	Design code to detect light. Putting code to pico board
Deliverables for each task	C++ code that automatically turns on the streetlight when motion is detected	Working C++ code that adjusts the streetlight brightness based on the light levels.	Light detection code, possible pico board, possible button.

Communication: We will communicate through the group chat and use GitHub to track our progress. If someone has trouble completing a task, we will discuss it as a team.

4. Document Your Plan

Project Scope: Our project is to create a smart streetlight controller using C++ that uses light levels and motion detection to decide when the streetlight should turn on, off, or change brightness.

Expected Deliverables: A fully functioning C++ program, GitHub repository with all the code, a final report, and a presentation.

Team Meeting Summary: We discussed how we wanted the smart streetlight controller to work and now have a general idea of what features we want to include. We decided to use C++ and GitHub for our project along with a Pico board. We also created a general timeline for the project.

Member	This iteration (completed)	Next iteration
Francisco	Helped create the project plan and discuss project objectives.	Have C++ code ready for the motion detection part of the controller.
Kenji	Helped create the project plan and organize the tasks and deadlines.	Have C++ code ready for the brightness control system.
Damilare	Helped create the project plan and decide milestones	Have C++ code ready for light detection part of controller